SQL Interview Questions

SQL Fundamentals

1. What is SQL, and why is it used?
2. What is a database?
3. What is a Database Management System (DBMS)?
4. What is a Relational Database Management System (RDBMS)?
5. What is the difference between DBMS and RDBMS?
6. What is a table in a database?
7. What are rows and columns in a table?
8. What is a SQL query?
9. What are the different categories of SQL commands?
10. What is the difference between DDL, DML, DQL, DCL, and TCL?
11. What is the purpose of the `SELECT` statement?
12. How do you retrieve all columns from a table?
13. What is the purpose of the `WHERE` clause?
14. What is the difference between `WHERE` and `HAVING`?
15. How do you retrieve unique values using `DISTINCT`?
16. What is the purpose of the `ORDER BY` clause?
17. What is the difference between ascending and descending order?
18. What is the purpose of the `LIMIT` clause in MySQL?
19. What are SQL aliases, and how do you use `AS`?
20. What is the difference between `DELETE`, `DROP`, and `TRUNCATE`?
21. What is the purpose of the `INSERT INTO` statement?
22. How do you modify existing records using `UPDATE`?
23. What happens if you execute an `UPDATE` statement without a `WHERE` clause?
24. What is the purpose of the `CREATE TABLE` statement?
25. How do you add a new column to an existing table?
26. What is the purpose of the `ALTER TABLE` statement?
27. What is the difference between `CHAR` and `VARCHAR`?

28. What are the common numeric, string, and date data types in MySQL?
29. What is the difference between `NULL`, zero, and an empty string?
30. What is the purpose of the `IS NULL` and `IS NOT NULL` operators?

Filtering and Constraints

31. What is the difference between the `WHERE` and `ORDER BY` clauses?
32. How do you filter records using comparison operators?
33. What are the logical operators `AND`, `OR`, and `NOT`?
34. What is the difference between `AND` and `OR`?
35. What is the purpose of the `BETWEEN` operator?
36. How does the `IN` operator work?
37. What is the purpose of the `LIKE` operator?
38. What is the difference between the `%` and `_` wildcards in SQL?
39. How do you search for values that begin with a specific character?
40. How do you search for values that contain a particular substring?
41. What is a primary key?
42. What is a foreign key?
43. What is the difference between a primary key and a unique key?
44. What is a composite primary key?
45. What is a candidate key?
46. What is a super key?
47. What is a surrogate key?
48. What are SQL constraints, and why are they important?
49. What is the purpose of the `NOT NULL` constraint?
50. What is the purpose of the `UNIQUE` constraint?
51. What is the purpose of the `DEFAULT` constraint?
52. What is the purpose of the `CHECK` constraint?
53. What is referential integrity?
54. What is the difference between `ON DELETE CASCADE` and `ON DELETE SET NULL`?
55. Can a table have multiple foreign keys? Explain with an example.

Aggregate Functions and Grouping

56. What are aggregate functions in SQL?
57. What is the difference between `COUNT(*)` and `COUNT(column_name)`?
58. What is the purpose of the `SUM()` function?
59. How does the `AVG()` function work?
60. What is the difference between `MIN()` and `MAX()`?
61. What is the purpose of the `GROUP BY` clause?
62. What happens when you use `GROUP BY` with multiple columns?
63. What is the purpose of the `HAVING` clause?
64. Why is `HAVING` commonly used with aggregate functions?
65. How do you find the total salary paid by each department?
66. How do you find departments with more than five employees?
67. How do you count the number of distinct values in a column?
68. How do aggregate functions handle `NULL` values?
69. What is the difference between filtering rows before grouping and filtering groups after aggregation?
70. How would you find the highest, lowest, and average salary for each department?

Joins and Relationships

71. What is a join in SQL?
72. Why are joins used in relational databases?
73. What is an `INNER JOIN`?
74. What is a `LEFT JOIN`?
75. What is a `RIGHT JOIN`?
76. What is a `CROSS JOIN`?
77. What is a self join?
78. What is the difference between `INNER JOIN` and `LEFT JOIN`?
79. What is the difference between `LEFT JOIN` and `RIGHT JOIN`?
80. What is the difference between a join condition in `ON` and a filter in `WHERE`?
81. What is an equi join?

82. What is a non-equi join?
83. What is a natural join, and why should it be used cautiously?
84. How do you join three or more tables in a single query?
85. How do you retrieve employees along with their department names?
86. How do you find customers who have never placed an order?
87. How do you find employees who earn more than their managers?
88. How can a self join be used to represent an employee-manager relationship?
89. What happens when join columns contain duplicate values?
90. How can you identify and prevent unexpected duplicate rows after joining tables?

Subqueries and Set Operations

91. What is a subquery in SQL?
92. What is the difference between a subquery and a join?
93. What is a single-row subquery?
94. What is a multiple-row subquery?
95. What is a correlated subquery?
96. What is the difference between correlated and non-correlated subqueries?
97. What is the purpose of the `EXISTS` operator?
98. What is the difference between `EXISTS` and `IN`?
99. What is the difference between `NOT EXISTS` and `NOT IN`, especially when `NULL` values are present?
100. What is a scalar subquery?
101. What is a derived table?
102. What is a Common Table Expression (CTE)?
103. What is the difference between a CTE and a subquery?
104. What is the difference between `UNION` and `UNION ALL`?
105. What are the requirements for combining query results using set operations?

Window Functions and Ranking

106. What is a window function in SQL?

107. How does a window function differ from an aggregate function used with `GROUP BY`?
108. What is the purpose of the `OVER()` clause?
109. What is the difference between `PARTITION BY` and `GROUP BY`?
110. What is the difference between `ROW_NUMBER()`, `RANK()` and `DENSE_RANK()`?
111. How do you find the second-highest salary using a window function?
112. How do you find the top three salaries in each department?
113. What is the purpose of the `LAG()` function?
114. What is the purpose of the `LEAD()` function?
115. How can you compare each month's sales with the previous month's sales?
116. What is a running total, and how can you calculate it?
117. What is a moving average in SQL?
118. What is the purpose of `FIRST_VALUE()` and `LAST_VALUE()`?
119. What is a window frame, and how do `ROWS` and `RANGE` differ?
120. How would you identify duplicate records using `ROW_NUMBER()`?

Database Design, Transactions and Performance

121. What is database normalization?
122. What is the difference between First Normal Form (1NF), Second Normal Form (2NF) and Third Normal Form (3NF)?
123. What is Boyce–Codd Normal Form (BCNF)?
124. What is denormalization, and when is it useful?
125. What is a functional dependency?
126. What is an index in SQL?
127. How does a database index improve query performance?
128. What is the difference between clustered and nonclustered indexes?
129. What is a composite index and why does column order matter?
130. What is a covering index?
131. What are the disadvantages of creating too many indexes?
132. What is a query execution plan?
133. How do you use `EXPLAIN` in MySQL?
134. What are transactions in SQL?
135. What are the ACID properties of a transaction?

136. What is the difference between `COMMIT` and `ROLLBACK`?
137. What is a savepoint, and how is it used?
138. What are dirty reads, non-repeatable reads and phantom reads?
139. What is the difference between optimistic and pessimistic locking?
140. What is a deadlock and how can it be prevented or handled?

MySQL Internals and Real-World Problem Solving

141. What is the difference between a stored procedure and a stored function?
142. What is a database trigger, and when should you use one?
143. What is a view, and how does it differ from a table?
144. What is a recursive CTE and where can it be useful?
145. How do you find gaps in a sequence of IDs or dates using SQL?
146. How would you optimize a slow query involving multiple joins and millions of records?
147. What is the difference between partitioning and sharding?
148. How does MySQL's InnoDB storage engine differ from the MyISAM storage engine?
149. How would you safely update millions of rows without causing excessive locking or downtime?
150. How would you design and query a database for an e-commerce system involving customers, products, orders, order items, payments and inventory?